Computer Networking

Chapter 1: Computer Networks and the Internet

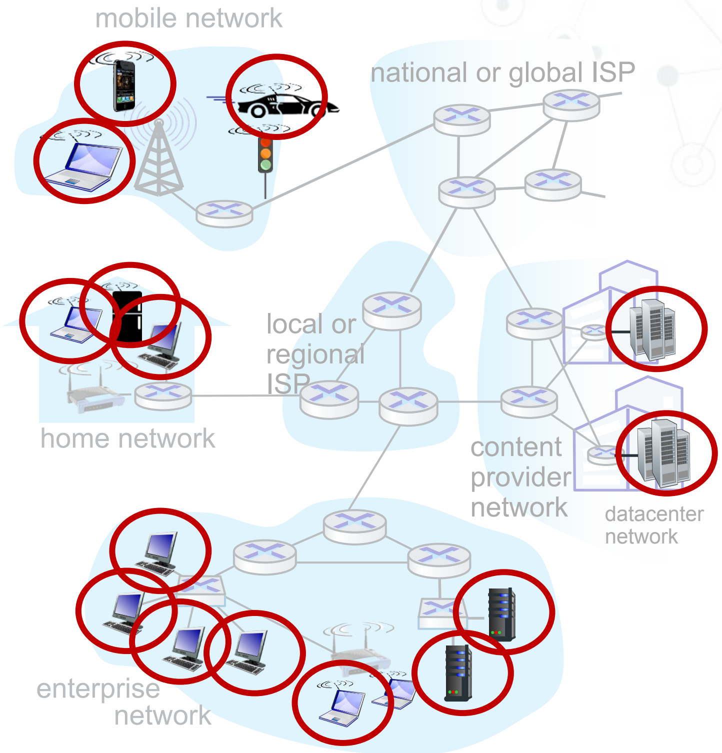
Overview/roadmap:

- What *is* the Internet?
- What *is* a protocol?
- The Network edge: hosts, access network, physical media
- The Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- Protocol layers, service models

A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers



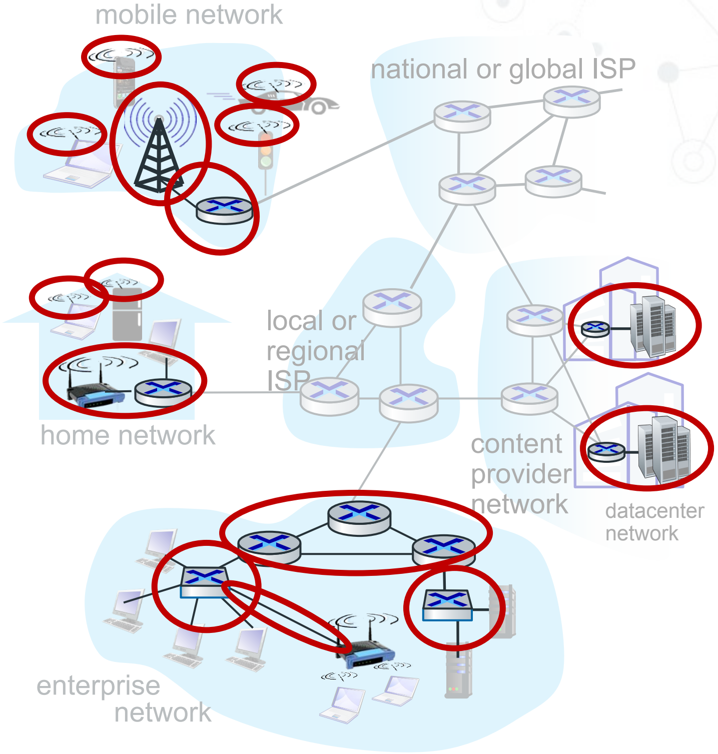
A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links
- Connects end systems to the internet

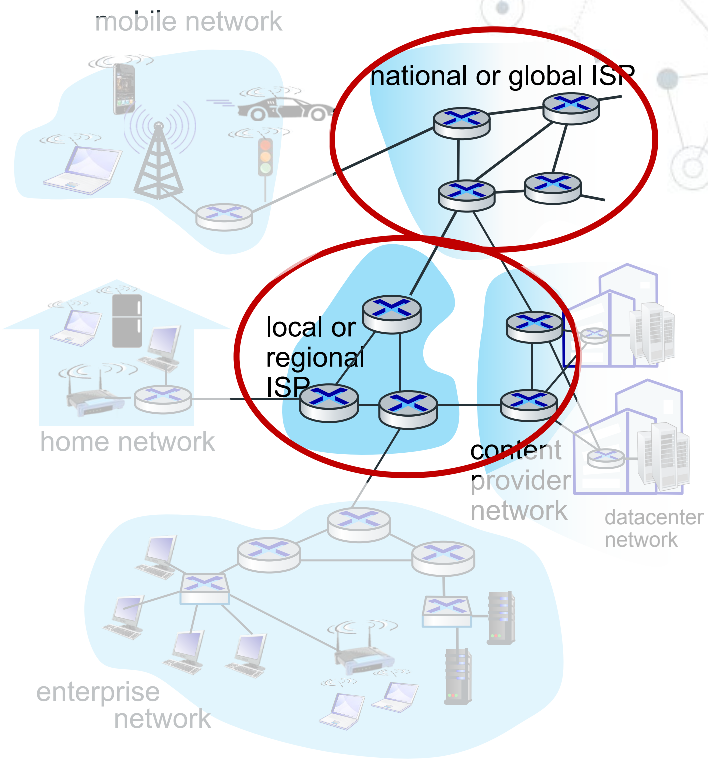


A closer look at Internet structure

Network edge:
- hosts: clients and servers
- servers often in data centers

Access networks, physical media:
- wired, wireless communication links

Network core:
- interconnected routers
- network of networks



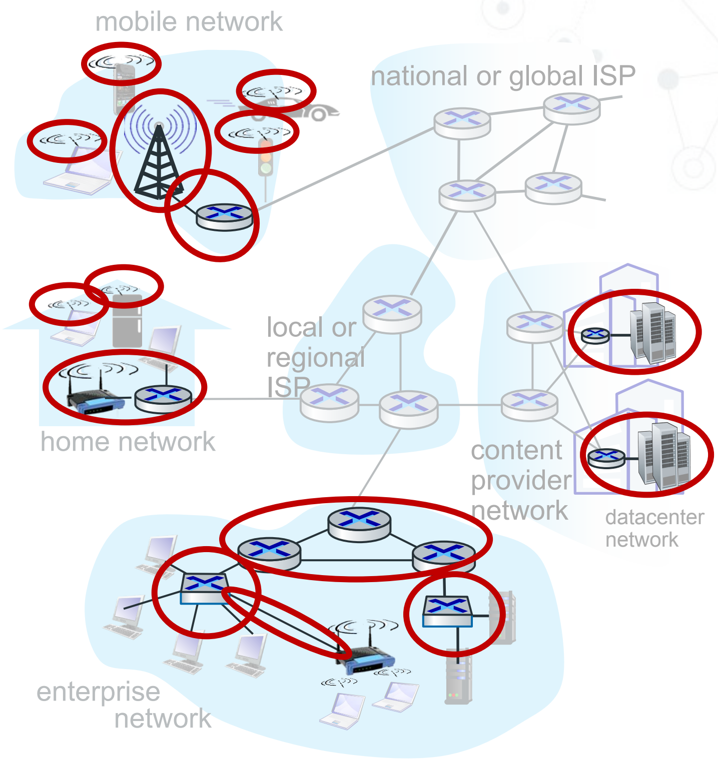
Access networks and physical media

Q: How to connect end systems to edge router?

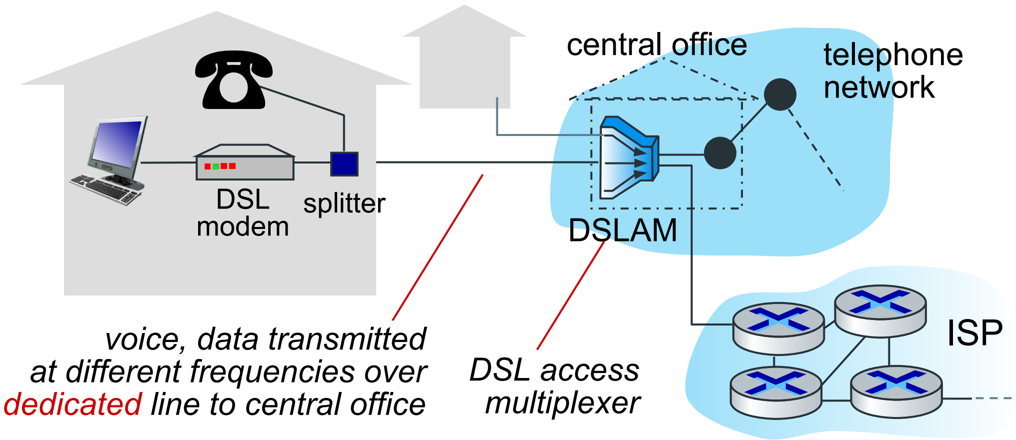
- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)

What to look for:

- transmission rate (bits per second) of access network?
- shared or dedicated access among users?



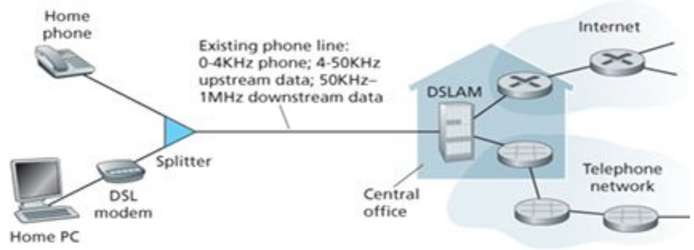
Access networks: digital subscriber line (DSL)



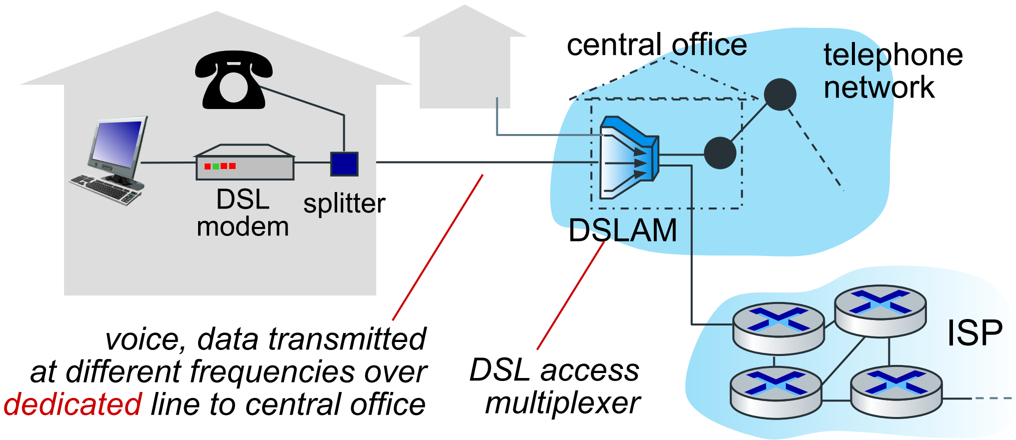
- One of the technology used for access N/W is DSL

Advantages:

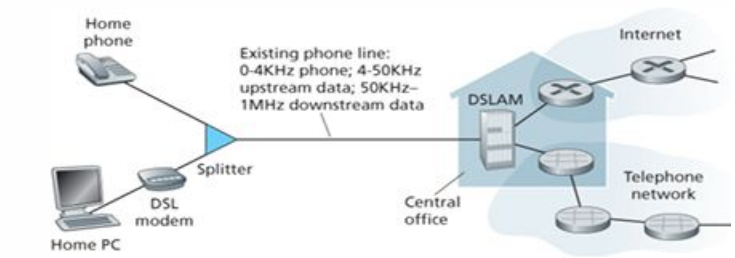
- Use existing telephone line to central office DSLAM
  - data over DSL phone line goes to Internet
  - voice over DSL phone line goes to telephone net
- 24-52 Mbps dedicated downstream transmission rate
- 3.5-16 Mbps dedicated upstream transmission rate



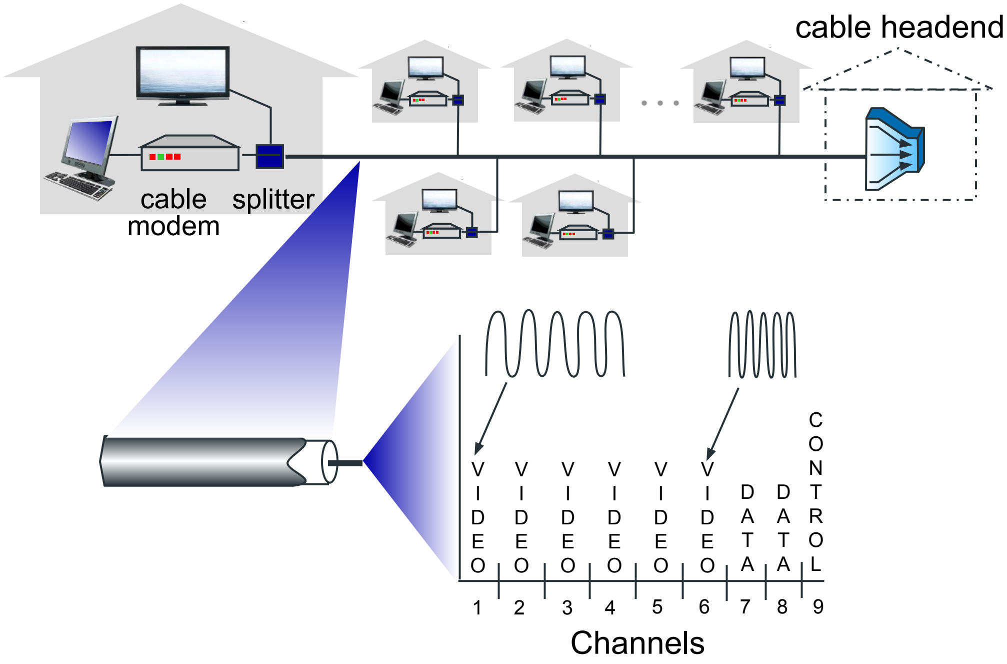
Access networks: digital subscriber line (DSL)



- The maximum rate is also limited by **the distance between the home and the CO**, the **gauge of the twisted-pair line** and the **degree of electrical interference**.
- Engineers have expressly designed DSL for short distances between the home and the CO; generally, if the residence is not located within 5 to 10 miles of the CO, the residence must resort to an alternative form of Internet access.

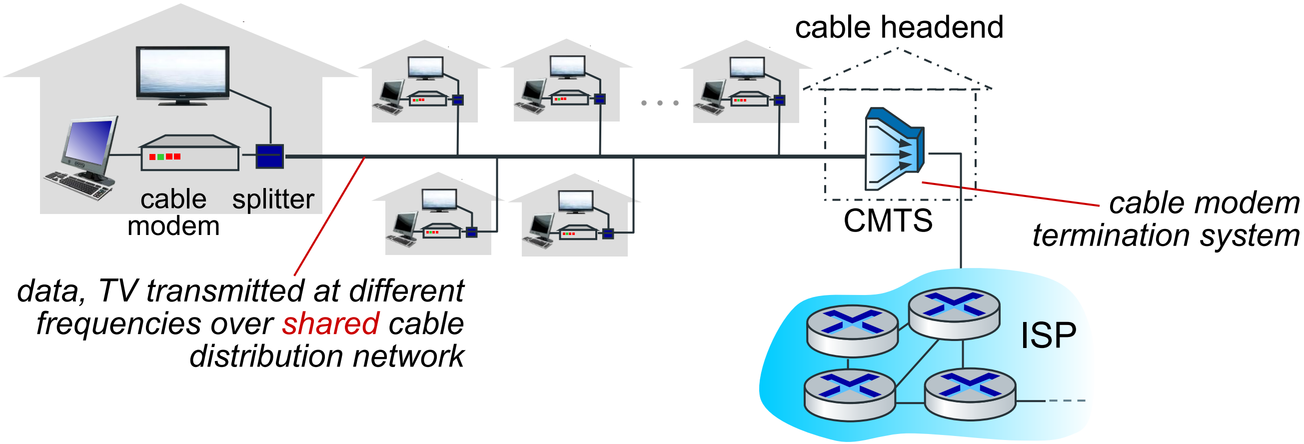


Access networks: cable-based access



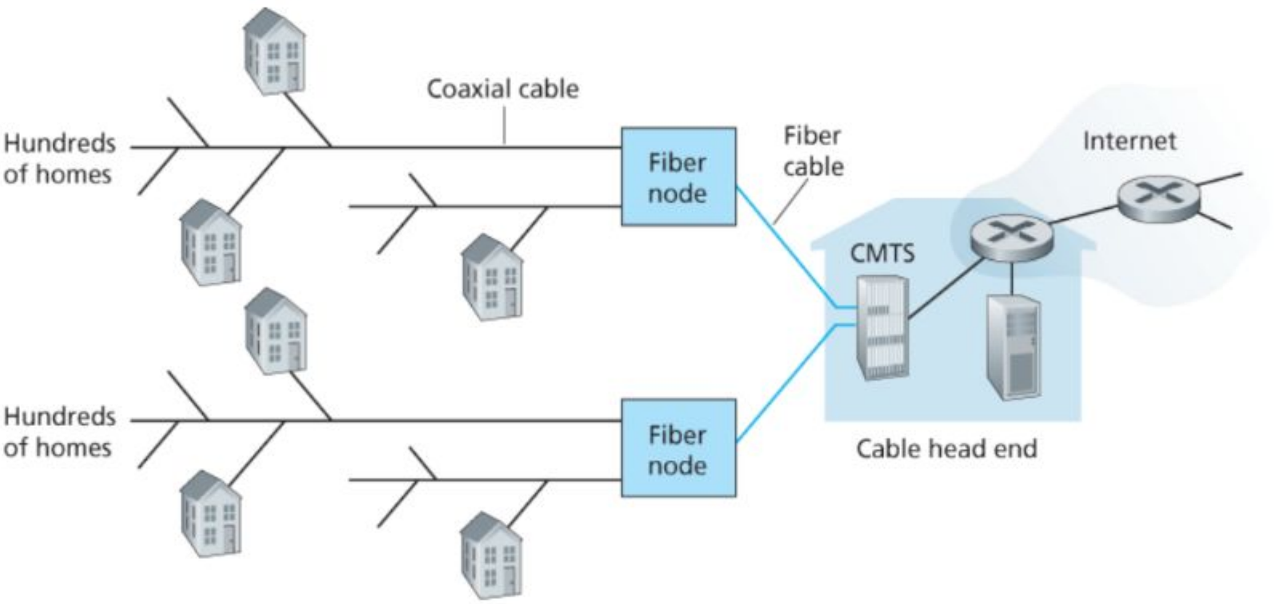
frequency division multiplexing (FDM): different channels transmitted in different frequency bands

Access networks: cable-based access



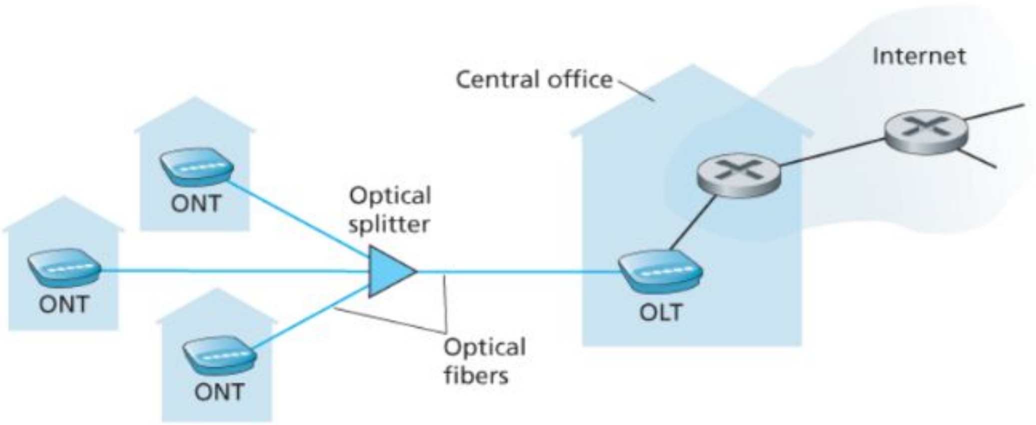
- HFC: hybrid fiber coax
  - asymmetric: up to 40 Mbps – 1.2 Gbs downstream transmission rate, 30-100 Mbps upstream transmission rate
- network of cable, fiber attaches homes to ISP router
  - homes *share access network* to cable headend

Access networks: cable-based access



A hybrid fiber-coaxial access network

Access networks: Fiber to the home internet access



- FTTH Internet access
- Optical splitter combines multiple homes on a single optical fiber
- Optical network terminator (ONT) provides internet to the home (optical to digital)
- Optical line terminator (OLT) providing conversion between optical and electrical signals, connects to the Internet via a telco router
- FTTH can potentially provide Internet access rates in the gigabits per second range

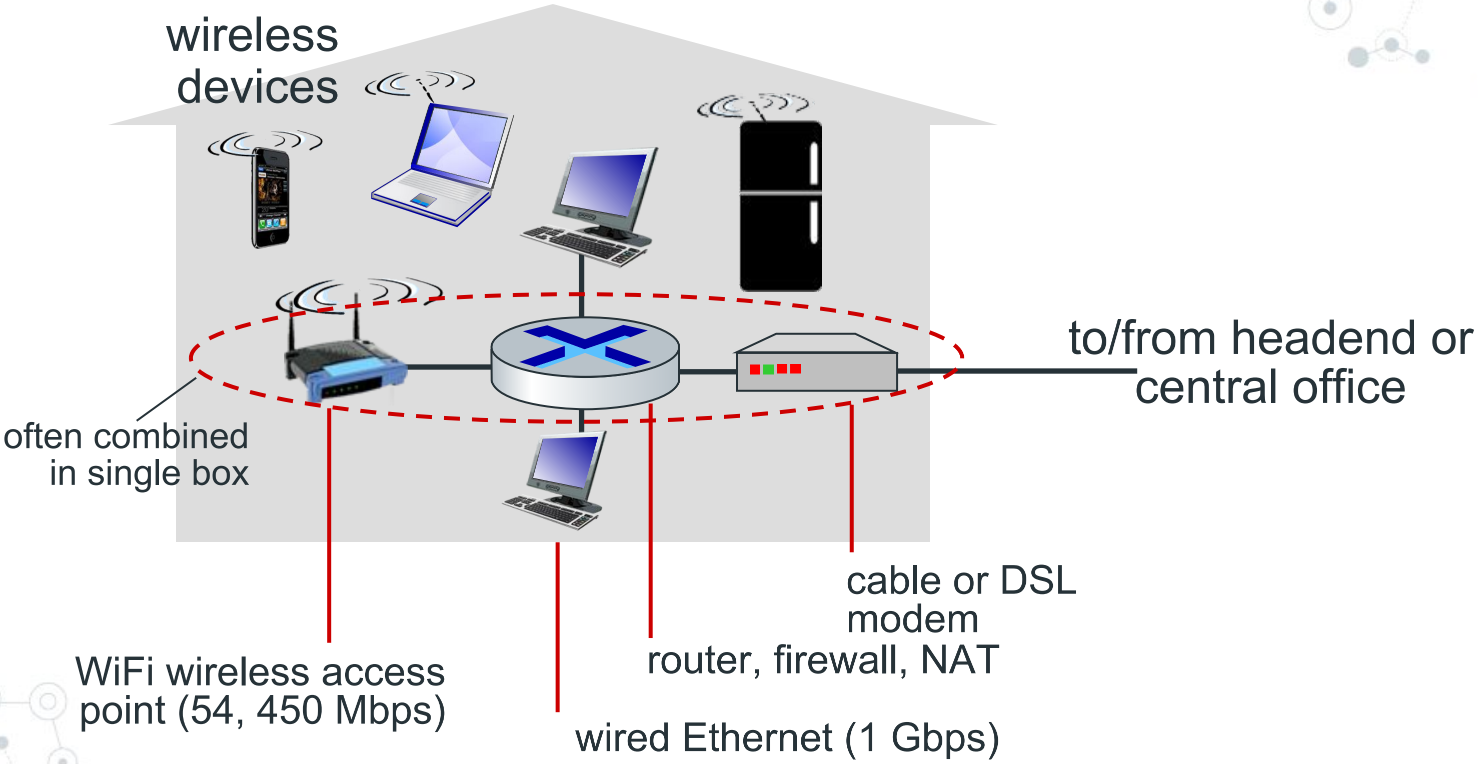
Access networks: Fiber to the home internet access

- Two other access network technologies are also used to provide Internet access to the home where DSL, cable, and FTTH are not available (e.g., in some rural settings),

    - A satellite link can be used to connect a residence to the Internet at speeds of more than 1 Mbps; StarBand and HughesNet are two such satellite access providers.

    - Dial-up access over traditional phone lines is based on the same model as DSL—a home modem connects over a phone line to a modem in the ISP. Compared with DSL and other broadband access networks, dial-up access is very slow at 56 kbps.

Access networks: home networks



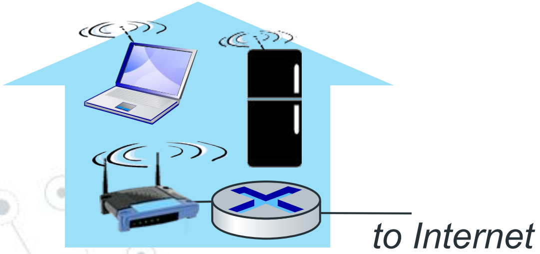
Wireless access networks

Shared *wireless* access network connects end system to router

- via base station aka “access point”

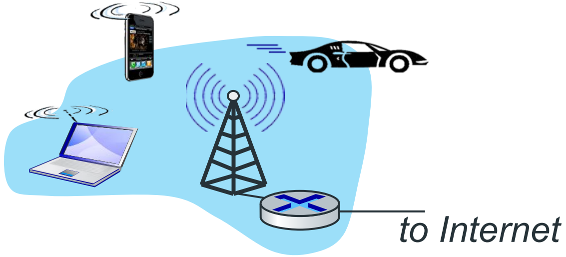
Wireless local area networks (WLANs)

- typically within or around building (~100 ft)
- 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rate

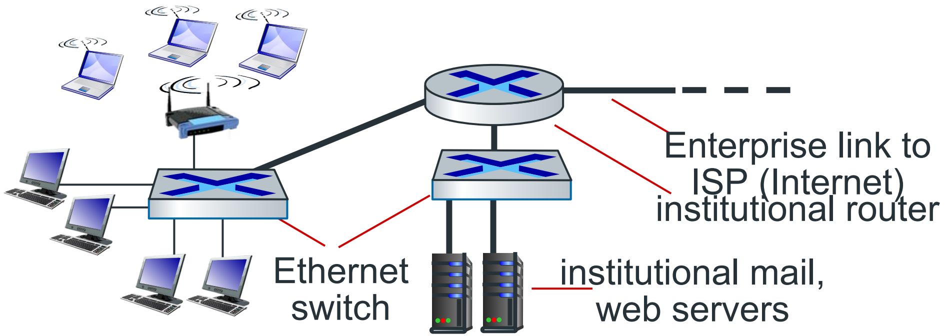


Wide-area cellular access networks

- provided by mobile, cellular network operator (10’s km)
- 10’s Mbps
- 4G cellular networks (5G coming)



Access networks: enterprise networks



- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers (we'll cover differences shortly)
  - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
  - WiFi: wireless access points at 11, 54, 450 Mbps

Links: physical media

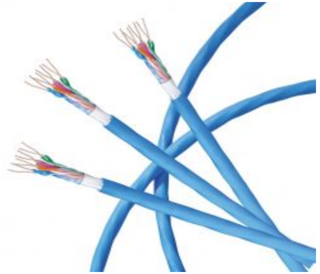
- Consider a bit traveling from one end system, through a series of links and routers, to another end system.
- The source end system first transmits the bit, and shortly thereafter the first router in the series receives the bit;
- The first router then transmits the bit, and shortly thereafter the second router receives the bit; and so on…
- From source to destination, it passes through a series of transmitter-receiver pairs.
- For each transmitter receiver pair, the bit is sent by propagating electromagnetic waves or optical pulses across a physical medium.
- The physical medium can twisted-pair copper wire, coaxial cable, multimode fiber-optic cable, and satellite radio spectrum.
- Physical media fall into two categories: **guided media and unguided media**.
  - With guided media, the waves are guided along a solid medium, such as a fiber-optic cable, a twisted-pair copper wire, or a coaxial cable.
  - With unguided media, the waves propagate in the atmosphere and in outer space, such as in a wireless LAN or a digital satellite channel

Links: physical media

- bit: propagates between transmitter/receiver pairs
- physical link: what lies between transmitter & receiver
- guided media:
  - signals propagate in solid media: copper, fiber, coax
- unguided media:
  - signals propagate freely, e.g., radio

Twisted pair (TP)

- two insulated copper wires
  - Category 5: 100 Mbps, 1 Gbps Ethernet
  - Category 6: 10Gbps Ethernet



Links: physical media

Coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
  - multiple frequency channels on cable
  - 100’s Mbps per channel

Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
  - high-speed point-to-point transmission (10’s-100’s Gbps)
- low error rate:
  - repeaters spaced far apart
  - immune to electromagnetic noise

Links: physical media

Wireless radio

- signal carried in electromagnetic spectrum
- no physical "wire"
- broadcast and "half-duplex" (sender to receiver)
- propagation environment effects:
  - reflection
  - obstruction by objects
  - interference

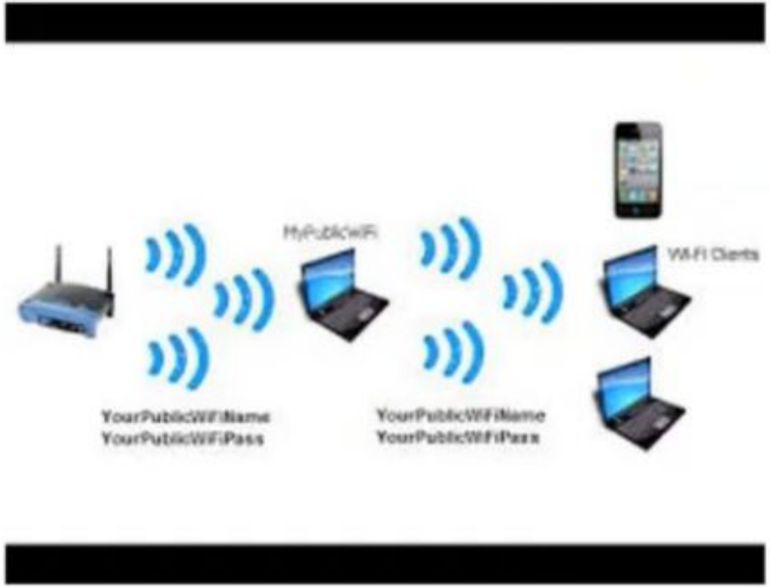
Radio link types:

- terrestrial microwave
  - up to 45 Mbps channels
- Wireless LAN (WiFi)
  - Up to 100's Mbps
- wide-area (e.g., cellular)
  - 4G cellular: ~ 10's Mbps
- satellite
  - up to 45 Mbps per channel
  - 270 msec end-end delay
  - geosynchronous versus low-earth-orbit

Links: physical media

LAN (e.g., WiFi)

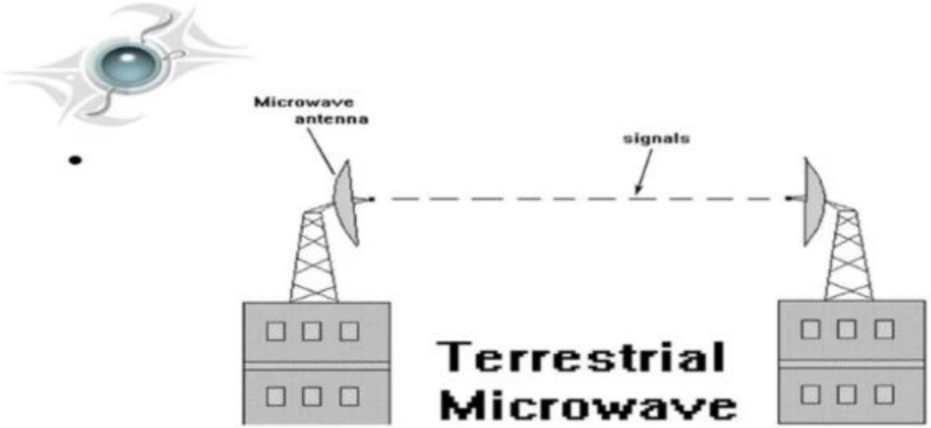
- 54 Mbps
- A building
- Penetrate through walls



Links: physical media

Microwave

- Used for long distance telephone service
- Parabolic dish mounted high
- Used for both voice and TV transmission



Links: physical media

Satellite

- Kbps to 45 Mbps channel (or multiple smaller channels
- 270 msec end-end delay